

Open Source Tools You Can Use to Handle Big Data

Big Data today influences our lives in the most unexpected ways, and organisations are using it extensively to gain that competitive edge in the market. So let's get acquainted with the open source tools that help us to handle Big Data.



Gone are the days when banks used to store customer information (such as names, photographs and specimen signatures) in individual postcard-like data sheets. That was an era where thick registers were used in different government offices like post offices, property tax collection centres, etc, to store customers' details or maintain the daily attendance records of employees. If an employee had to update any of the registered customer's details, the task could take up the whole day. Hours were wasted searching for that particular customer's details and then creating a new record to replace the old one. The customers, too, had to wait for hours for such minor tasks to be completed. Apart from the tediousness of searching for data from piles of ledgers, such paper files could be lost at any time due to disasters like floods or fire, apart from the degradation of the very paper on which the data was recorded.

Today, on our journey towards a digital India, all government offices are switching to digitisation instead of manual record keeping. As we proceed along this path, we

see a tremendous increase in the size of data. There are around 230 billion tweets posted on Twitter per day, 2.7 billion Likes and comments added to Facebook daily, and around 60 hours of video uploaded to YouTube every minute. All this leads to about 2.5 exabytes of data being generated on a daily basis by different online applications, transactional data sources, IoT devices, etc. The term that encapsulates such immense volumes of information is Big Data. Existing hardware and software systems are unable to handle such volumes of different types of data being created at such enormous speed. The data has also become too complex and dynamic to be stored, processed, analysed and managed with traditional data tools. So Big Data is now analysed by computer systems to reveal specific trends, patterns and associations, especially those relating to human behaviour and interactions. We are making our machines smart enough to 'think and decide' on what action needs to be performed and when by using Big Data techniques like predictive analytics, user-behaviour analytics, etc.